

Smart Attendance System

Using Facial Recognition

An Android-based approach to fast, accurate and accountable attendance management.

Live camera → Face match → Attendance record

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02 Objectives

What the solution is designed to achieve

1 Automate attendance

Replace repetitive manual marking with face-based verification.

2 Recognize in real time

Detect and identify registered users through a live camera.

3 Record accurately

Store the verified person, date and time automatically.

4 Reduce errors

Minimize missed entries, duplicate work and proxy attendance.

5 Secure & synchronize

Maintain user and attendance records in a centralized database.

6 Enable administration

Support user management, monitoring, reports and analytics.

DESIGN GOAL **Efficient • Accurate • Scalable**

03 Problem Statement

Manual attendance creates operational friction and weakens record reliability

THE CURRENT REALITY

A simple daily task becomes a recurring administrative burden.

The challenge grows with every additional class, employee, register and reporting period.

Time

Calling names and entering data delays sessions.

Accuracy

Human error produces missing or inconsistent records.

Scale

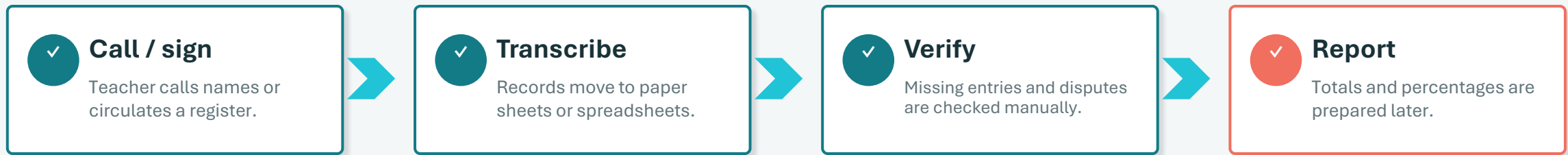
Large groups make verification and follow-up difficult.

Control

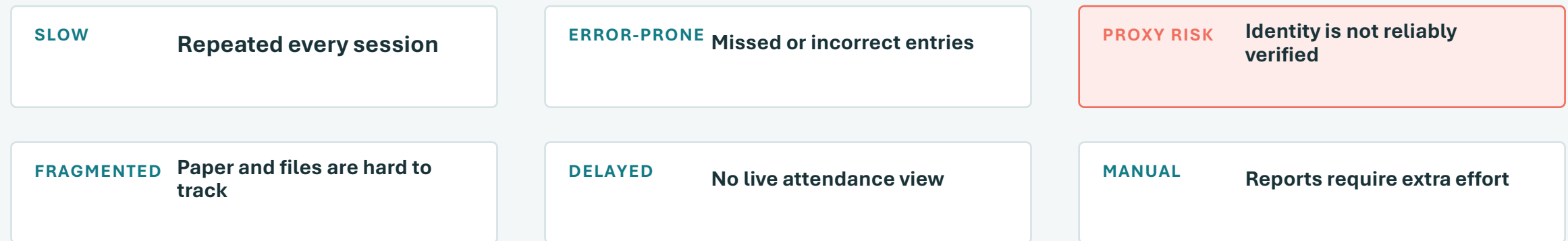
Paper and spreadsheets limit monitoring and auditability.

04 Existing System

Typical manual workflow and its practical limitations



Where the process breaks down



05 Proposed System

Facial verification turns attendance into a controlled digital event

01

REGISTER

Create a profile and capture facial data in advance.

02

RECOGNIZE

Match the live face with a registered identity.

03

RECORD

Write the attendance entry with date and time.

04

REVIEW

Monitor records and generate reports from the dashboard.

EXPECTED OPERATIONAL CHANGE

Manual marking → **Automatic verification**

Delayed entry → **Real-time recording**

Paper / sheets → **Central database**

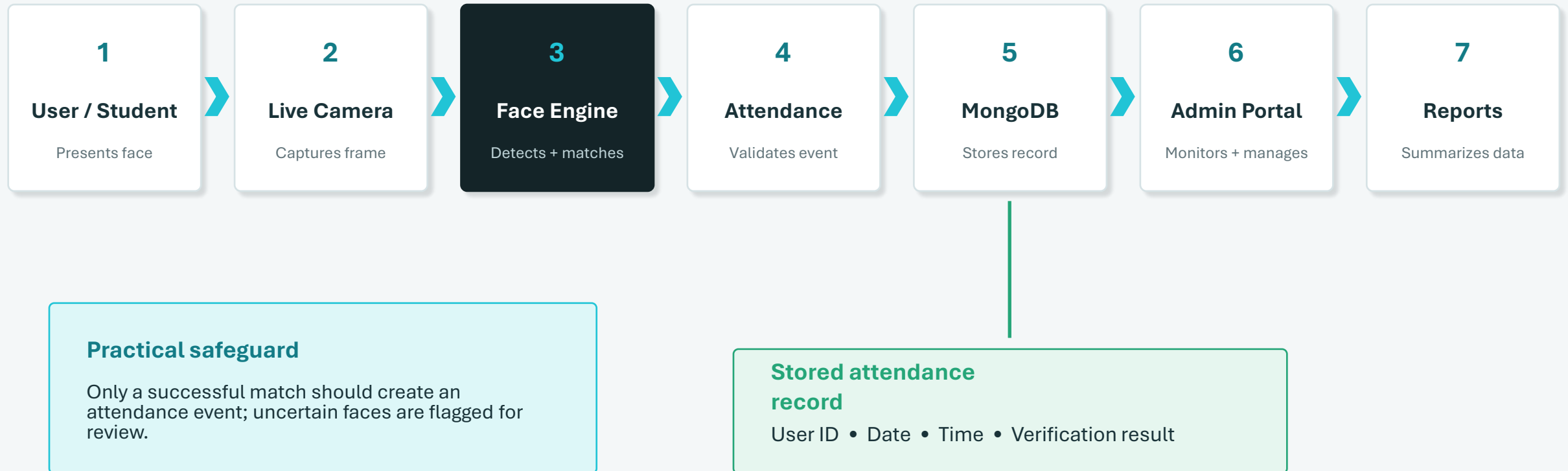
Manual totals → **Automated reports**

Limited visibility → **Live administrator view**

Faster • More accurate • Easier to audit

06 System Architecture

End-to-end data flow from camera capture to administrative insight



07 Technology Stack

A practical web-and-vision stack for the attendance workflow

INTERFACE

HTML • CSS • JavaScript

User-facing attendance and administration screens.

BACKEND

Python • Flask

Application logic, requests and service coordination.

FACE AI

OpenCV • Face Recognition

Detection, encoding comparison and identification.

DATABASE

MongoDB

Student profiles, face identifiers and attendance records.

CONNECTIVITY

PyMongo

Database access and record operations from Flask.

HARDWARE

Webcam / CCTV

Real-time image capture at the attendance point.

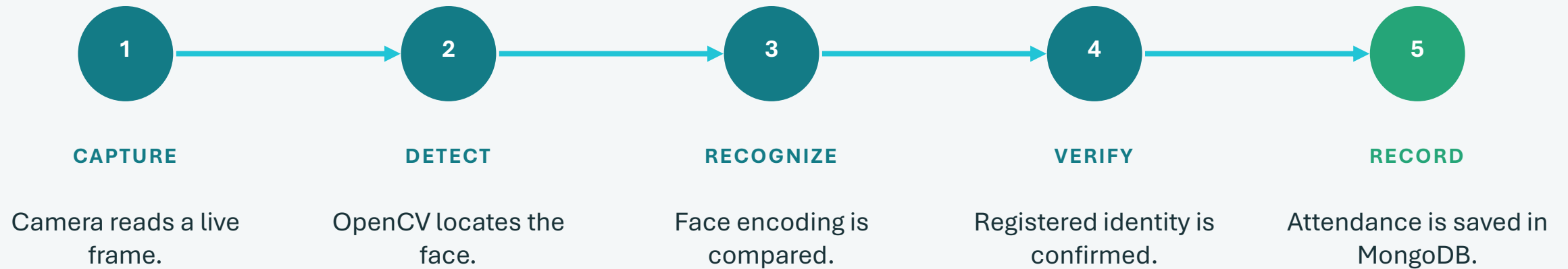
DEVELOPMENT

Visual Studio Code

Coding, testing and project development.

08 Working of the System

The recognition session follows a controlled, repeatable sequence



If matched Create one timestamped attendance entry and show confirmation.
Duplicate checks can prevent repeated marking in the same session.

If uncertain Do not mark automatically;
request another scan or administrator review.

09 Key Features

Core capabilities mapped to everyday attendance operations

LIVE

Real-time recognition

Identify registered users from a live camera feed.

AUTO

Automatic marking

Create attendance without manual name calling.

TIME

Date & time

Keep a timestamp for every verified entry.

DATA

Central records

Store profiles and attendance in one database.

ADMIN

Management dashboard

Manage students and review attendance status.

REPORT

Reports & analytics

Review records and attendance percentages.

Practical result: less time spent marking attendance, fewer manual errors, and clearer administrative visibility.

Thank You

Questions & Discussion

A practical step toward faster, more reliable and accountable attendance management.

Camera → Recognition → Verification → Record

